COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212-Automating Server Management	**SECTION:**	CPE31S2
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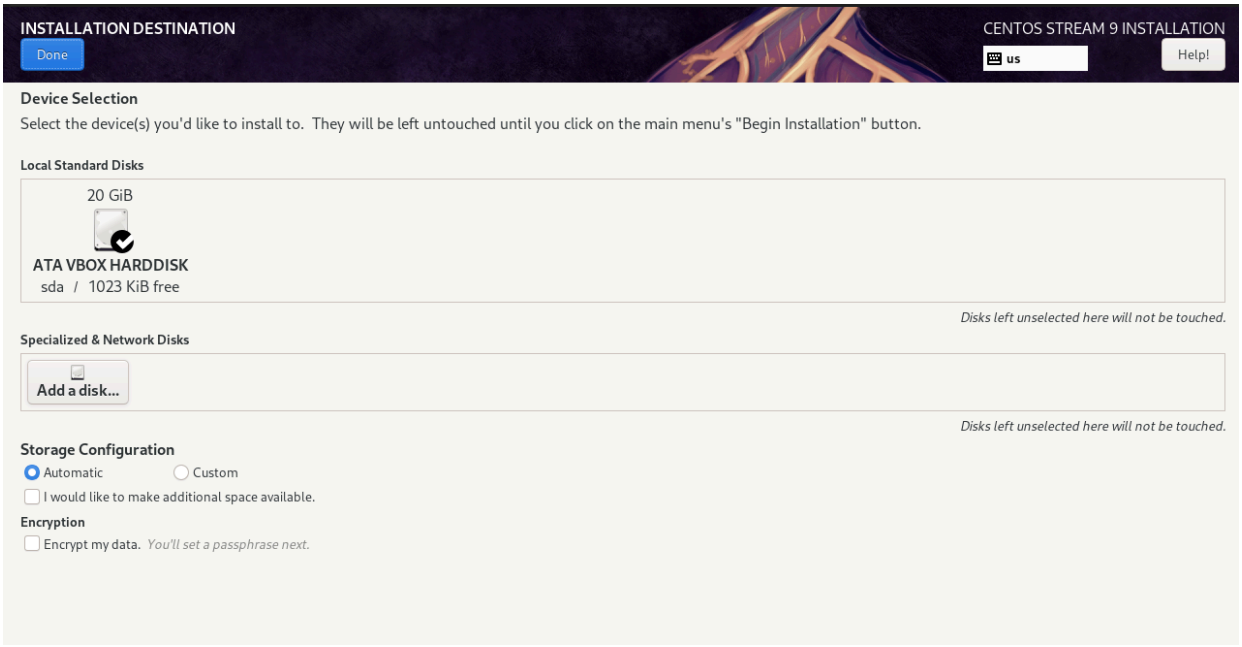
3.1 Hands-on Activity Install SSH server on CentOS or RHEL 8

1. PROCEDURE OUTPUT

What procedures did you perform?

Task 1: Download the CentOS or RHEL-8 image

2. Create a VM machine with 2 Gb RAM and 20 Gb HD.



3. Install the downloaded image.

4. Show evidence that the OS was installed already.

Task 2: Install the SSH server package openssh

1. Install the ssh server package openssh by using the dnf command:

$ dnf install openssh-server

```
[cent-server@localhost ~]$ sudo dnf install openssh-server
Last metadata expiration check: 0:09:21 ago on Fri 28 Aug 2026 01:11:24 PM PST.
Package openssh-server-9.9p1-13.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
```

2. Start the sshd daemon and set to start after reboot:

$ systemctl start sshd

```
[cent-server@localhost ~]$ systemctl start sshd
==== AUTHENTICATING FOR org.freedesktop.systemd1.manage-units ====
Authentication is required to start 'sshd.service'.
Authenticating as: cent-server
Password:
==== AUTHENTICATION COMPLETE ====
```

$ systemctl enable sshd

```
[cent-server@localhost ~]$ systemctl enable sshd
==== AUTHENTICATING FOR org.freedesktop.systemd1.manage-unit-files ====
Authentication is required to manage system service or unit files.
Authenticating as: cent-server
Password:
==== AUTHENTICATION COMPLETE ====
[cent-server@localhost ~]$
```

3. Confirm that the sshd daemon is up and running:

$ systemctl status sshd

```
[cent-server@localhost ~]$ systemctl status sshd
● sshd.service - OpenSSH server daemon
     Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enabled)
     Active: active (running) since Fri 2026-08-28 12:31:44 PST; 51min ago
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 865 (sshd)
      Tasks: 1 (limit: 20972)
     Memory: 2.1M (peak: 2.3M)
        CPU: 33ms
     CGroup: /system.slice/sshd.service
             └─865 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Aug 28 12:31:44 localhost.localdomain systemd[1]: Starting OpenSSH server daemon...
Aug 28 12:31:44 localhost.localdomain sshd[865]: Server listening on 0.0.0.0 port 22.
Aug 28 12:31:44 localhost.localdomain sshd[865]: Server listening on :: port 22.
Aug 28 12:31:44 localhost.localdomain systemd[1]: Started OpenSSH server daemon.
[cent-server@localhost ~]$
```

4. Open the SSH port 22 to allow incoming traffic:

$ firewall-cmd --zone=public --permanent --add-service=ssh

$ firewall-cmd --reload

```
[cent-server@localhost ~]$ sudo firewall-cmd --zone=public --permanent --add-service=ssh
[sudo] password for cent-server:
Warning: ALREADY_ENABLED: ssh
success
[cent-server@localhost ~]$ firewall-cmd --reload
Authorization failed.
    Make sure polkit agent is running or run the application as superuser.
[cent-server@localhost ~]$ sudo firewall-cmd --reload
success
[cent-server@localhost ~]$
```

5. Locate the ssh server man config file /etc/ssh/sshd_config and perform custom configuration. Every time you make any change to the /etc/ssh/sshd-config configuration file reload the sshd service to apply changes:

$ systemctl reload sshd

```
[cent-server@localhost ~]$ systemctl reload sshd
==== AUTHENTICATING FOR org.freedesktop.systemd1.manage-units ====
Authentication is required to reload 'sshd.service'.
Authenticating as: cent-server
Password:
==== AUTHENTICATION COMPLETE ====
[cent-server@localhost ~]$
```

Task 3: Copy the Public Key to CentOS

1. Make sure that ssh is installed on the local machine.

2. Using the command ssh-copy-id, connect your local machine to CentOS.

```
(base) nikko@Nikko:~$ ssh-copy-id cent-server@192.168.56.106
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/nikko/.ssh/id_rsa.pub"
The authenticity of host '192.168.56.106 (192.168.56.106)' can't be established.
ED25519 key fingerprint is SHA256:bsGVCKg4jEr/PI+nrh6UGTZF6iVy3dKWs2SO90Wlefk.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
cent-server@192.168.56.106's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'cent-server@192.168.56.106'"
and check to make sure that only the key(s) you wanted were added.
```

3. On CentOS, verify that you have the authorized_keys

```
[cent-server@localhost ~]$ cat ~/.ssh/authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAACAQDXWSxLkw92AGtZhOtKTujHhOh5L9TJXYhGP4oIsjhTuK1/PAsvba/e6hzfm81+kE5ys/uqA9F13kPbjqQT
G1C1ThRXRiX3441CC2Ac4OxAl29IyM/evFraptB7W1HUB8QN9iFBG3ShVsLUMffddfwWWeE6vLjgJoQuKdgPV8vQwmfZiU8pE64/XSXfHyymqlvXwpjqOFuQ
cDKHAp8d/tinXQiFF08uAGbXsLxHZ9SM0lCDQg12MDiY7qXBXMsn8YNpWnUB/OE6HnAL8uxH+79v2DzCvgM9I5V+T4d/zQYn2+p5YJcQMqrm6UXR064yY6WD
BqnBE7HRajcG1+iz1adYS81+JleN+Zf1h8v5HvS4qxThkNgZHsXjDTLq/tnELpIzxyrNP2LYY9teTgdNa41FtHBSpf+3de+QHrGzRjFVptVPDR3wweQHPHWN
33NPaDjytKvV01TpavskAMWtFPcKmFPFa81kYa4ZQfd+tDwArmyGp3PVwVVBzQor0yD6FCLlwtibbzJpSw== nikko@Nikko
[cent-server@localhost ~]$
```

Task 4: Verify ssh remote connection

1. Using your local machine, connect to CentOS using ssh.
2. Show evidence that you are connected

```
(base) nikko@Nikko:~$ ssh cent-server@cent-server
The authenticity of host 'cent-server (192.168.56.106)' can't be established.
ED25519 key fingerprint is SHA256:bsGVCKg4jEr/PI+nrh6UGTZF6iVy3dKWs2SO90Wlefk.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:19: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'cent-server' (ED25519) to the list of known hosts.
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Fri Aug 28 13:26:07 2026
[cent-server@localhost ~]$
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

Answer the following:

1. What do you think we should look for in choosing the best distribution between Debian and Red Hat Linux distributions?

- I think when choosing a distro one should consider first if the hardware architecture is supported by that specific distro and the package manager familiarity since I had a hard time using this distro because of the package manager.

2. What are the main differences between Debian and Red Hat Linux distributions?

- The difference is that CentOS is a downstream rebuild of the commercial Red Hat distribution, whereas Debian is a free upstream base distribution.

Conclusion:

- In conclusion, I've learned how to use other package manager like dnf since I need it to be able to use centOS or other RHEL distros. I also noticed that the centOS use sshd not ssh which gave me a hard time and a lot of errors and now I also consider a lot of advantages and disadvantages of a distro before downloading or using them for personal use. To sum this up this activity is just like 1 and 2 but with different distro to use.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini and grammarly**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Grammar correction, explanation of concepts**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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